# UrbanHeat AI

## Al Khor area study, Qatar

Theme: Urban Expansion, Land Use Change & Heat Risk

Satellite and local evidence for heat-protection site assessments

Team Al Zubarah, Qatar

UAE Space Academy hackathon, October 2026

# The planning problem

Proposed user: a municipal planning team choosing where to inspect first.

Shortlist cells, then inspect use, shade, access, ownership and feasibility before choosing an intervention.

Proposed value: organise evidence and inspection decisions. No adoption or measured benefit is established.

The prototype supports investigation; it does not establish heat-health risk.

# Bus-stop site assessment: V04-23

Satellite: 50.4 °C mean of two cell medians; 0% detected green pixels at the current threshold.

Local evidence: roadside trees, no shelter and limited walking access; Nasser confirms site visits.

Next check: waiting-time shade, passenger use, route safety, ownership and feasibility.

300 m September values are not stop measurements. October field photos show context, not measured use or shade.

# Open satellite data and mapped context

Sentinel-2: green-pixel indicators from 5, 15 and 30 September 2026.

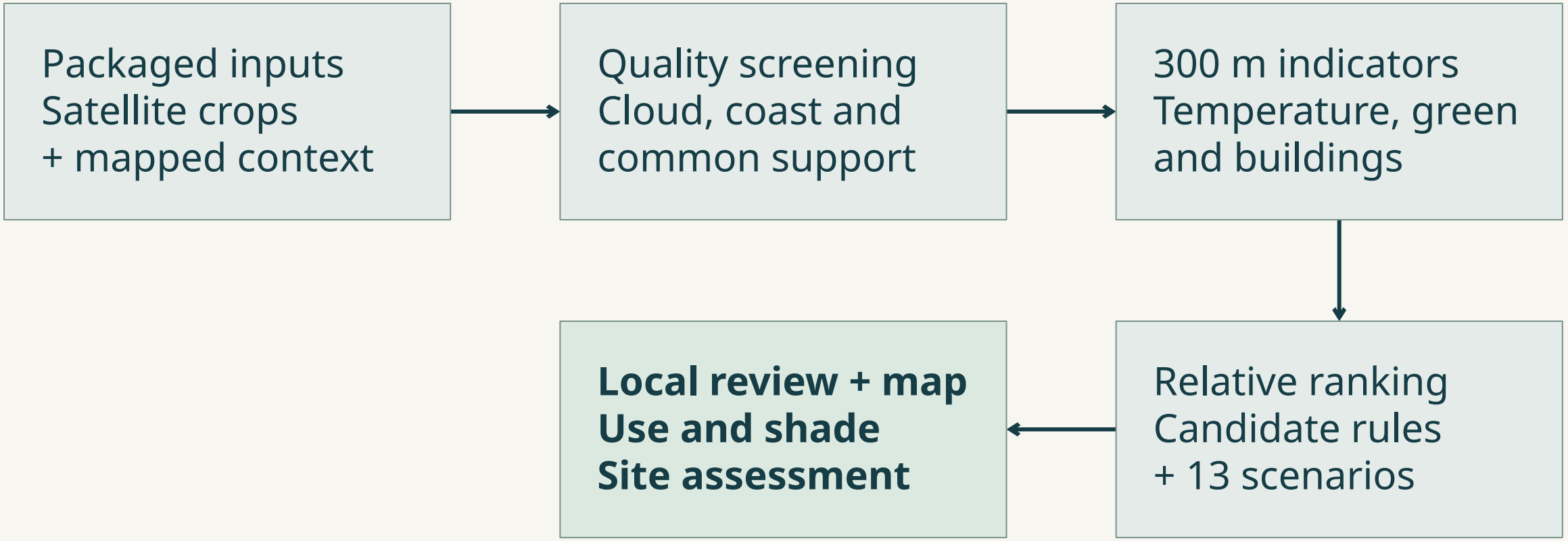
Landsat: surface temperature from 14 and 30 September. The 6 September scene fails coverage QA.

WorldCover 2021: historical land-cover context. OpenStreetMap: contributed building footprints downloaded 5 October.

Field photos: 8 October 2026; thermal data: September. No hyperspectral, 813 or proprietary VHR data.

Thermal native detail is about 100 m, delivered on a 30 m grid. We report 300 m cells.

# From source crops to an inspectable shortlist




External AI land cover: WorldCover for heat screening; Dynamic World in a separate experiment.
QA/ranking are rule based. No new model trained.

# Al Khor presentation examples

## V04-23 | Bus-stop road



Trees and stop sign; no shelter visible.

## V28-17 | Stadium parking



Stadium and parking; use unmeasured.

## V18-26 | Jogging route



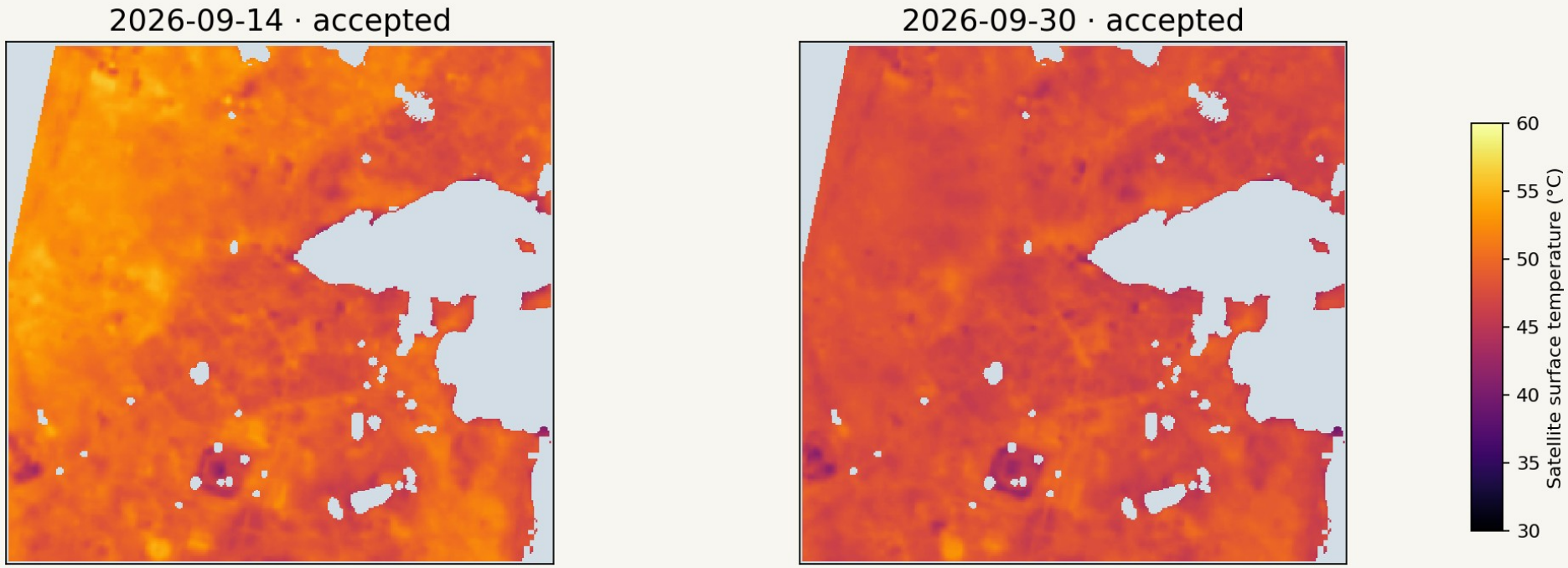
Marked path beside road and sandy ground.

8 October 2026 | 13:00–14:00 Qatar (UTC+3) | Shared capture window

Photographer: Abdulrahman Almohannadi | Confirmed by Nasser, not EXIF | Exact per-file time: Unknown

Team-selected examples; all 1,111 observed cells and 373 candidates remain. The satellite top three are unchanged.

# Matched surface-temperature observations




6 September excluded: 0% of historical urban inland support passed the strict filters.
94.4379 km² of common inland sample footprints. Grey: excluded or missing observations.

Morning surface observations, not air temperature. Native thermal detail about 100 m, delivered at 30 m.
Date differences do not establish long-term cooling. Landsat imagery courtesy of USGS.

94.4379 km² of common inland sample footprints across the broad study window.

# Three examples for site investigation

| Case study | Surface °C* | Green %* | Buildings % | Rank range |
|---|---|---|---|---|
| V04-23 | 50.4 | 0.0 | 2.3 | 1-14 |
| V28-17 | 50.0 | 0.0 | 0.1 | 1-23 |
| V18-26 | 47.47 | 0.0386 | 0.0 | 203–336 |

Bus stop: waiting-area shade. Stadium: event use. Jogging route: reported lack of path shade. Photos received; use and detailed shade unmeasured.

Team-selected examples, not the top three ranks. V18-26 ranks 275/373; hospital land stays in the full results. Visits do not measure use or shade.

*Means of date-level cell summaries. V18-26: 48.02/46.92°C; 72% coverage. Scenario ranges are not confidence intervals.

# Reproducibility and reference checks

The package includes 35 hashed input files, source metadata and pinned Python 3.12 dependencies.

Clean GitHub execution passed through a normal Jupyter kernel: five cells, zero errors and all 35 input hashes verified.

NDVI >=0.30 at 24/25 grass references and 0/30 building references. All 30 coastal references are excluded from retained thermal support.

Contributed reference labels have conflicts. Human confirmation and independent thermal validation remain pending.

# Potential impact and incubation next steps

Completed: satellite screening, sensitivity checks, local examples and preliminary review of 18 sites across four dates.

Proposed pilot: planner feedback and field shade/use/access checks; evaluate shortlist usefulness and feasibility.

Limited Dynamic World experiment: common coverage 30.94% (50% gate). Independent scientific validation pending; 2023 deferred.

The current PoC does not measure urban expansion, health risk, causal cooling or long-term warming.

# Team responsibilities

Nasser Alhuri: team coordination, site-visit confirmation and submission.

Abdulrahman Almohannadi: supplied 11 photos + one video; project review remains pending.

Mohammed Almarri: rehearse the live pitch (up to 10 min) and 5 min Q&A.

Majed Alkuwari: review slides and explain the limitations.

Ali Alkubaisi: rehearse the offline demo and prepare five-minute judge Q&A.

Photographed-site visits and supplied media collection confirmed by Nasser. Detailed assessment, member reviews and rehearsals remain pending.

# Data attribution and sources

Actual sources: Copernicus Sentinel, Landsat (USGS), WorldCover 2021, OSM; separate experimental Dynamic World exports (2021, 2026).

Credits: modified Copernicus Sentinel data (2026); Landsat courtesy USGS; WorldCover CC BY 4.0; OSM ODbL.

Dynamic World, CC BY 4.0: This dataset is produced for the Dynamic World Project by Google in partnership with National Geographic Society and the World Resources Institute.

Field photos: Abdulrahman Almohannadi, 8 October 2026, shared 13:00–14:00 Qatar (UTC+3) window. Confirmed by Nasser, not EXIF; exact per-file times Unknown.

Modified Copernicus Sentinel data (2021, 2026). Full sources, licences and experimental limits accompany the package. Code: MIT.